

Blockchain-Based POAP Attendance Badge System for Event Attendance Verification

Tan Jin Yi ^{a)}, Savithar Ravichenderan ^{b)}, Nurul Izzati Afina Sukri ^{c)}, and Muhammad Ameer Rafiqi Mohammad Shahizam ^{d)}

Author Affiliations

Faculty of Computing and Informatics, Multimedia University 63100, Cyberjaya, Selangor, Malaysia

Author Emails

^{a)} 1221305792@student.mmu.edu.my

^{b)} 1211111607@student.mmu.edu.my

^{c)} 1091106071@student.mmu.edu.my

^{d)} 1211106255@student.mmu.edu.my

Abstract. Event attendance systems are usually based on the use of paper sign-in sheets, QR codes or centralized databases. Even though such approaches work well, they still remain subject to various shortcomings including fake participation claims, duplications, unauthorized data editing, and poor event attendance verification. This paper presents the implementation of a Proof-of-Attendance Protocol (POAP) attendance badge system for event attendance verification using Ethereum smart contracts. The aim of the proposed solution is to allow organizers to mint tokens called attendance badges, which serve as proof of participation in an event. This work has been implemented via Solidity, ERC-1155, Hardhat, MetaMask, Sepolia testnet, and a basic frontend interface. ERC-1155 token standard has been used since it enables issuing different types of attendance badges through the same smart contract. The system is capable of creating events, managing issuer roles, generating QR-based check-in links for badge claiming, minting individual tokens and tokens in batches, managing the status of events, and badge verification of participants. Functional testing, gas cost analysis, code coverage analysis, and Sepolia testnet deployment tests have been done to evaluate the system. The results demonstrate successful implementation of all 15 automated test groups with 100% line and statement coverage of the smart contract.

INTRODUCTION

Verifying attendance is a crucial procedure in seminars, workshops, conferences, academic programs and professional training events. Most of the usual methods of attendance include the use of paper sign-in sheets, QR code scanners, or centralised platforms. These are simple methods but require a central point of data collection and management for attendance. This establishes a trust problem as centralized attendance data may be edited, deleted, duplicated, or lost if the system isn't well protected. It's possible that the participant could falsely claim attendance and that the organiser may struggle to establish the authenticity of attendance after the event has already taken place.

Attendance can be recorded and provided as proof of participation, success, and/or training completion in academic and workplace settings. Thus, it is important that the record be clear, verifiable, and tamper-proof. Blockchain technology has relevance in this problem since it provides decentralization, transparency, immutability and programmable execution via smart contracts. After confirming a blockchain transaction, it is hard to change without leaving a clue. Smart contracts can also be used to automatically follow system rules, e.g., who can add an event badge, who can add an attendance badge.

Proof-of-Attendance Protocol (POAP) is a blockchain-based protocol that provides digital badges or tokens as proof of attendance at an event. Rather than using a centralized attendance record system, participants have digital badges in their blockchain wallets. Later, these badges can be checked via blockchain records. The goal of this project is to create and deploy a blockchain-based system to verify attendance at an event (POAP). The ERC-1155 token standard is utilized for multiple types of event badges in a single smart contract. It also supports owner control, role management for issuers, single badge issuance, batch badge issuance, event status control, and attendee badge verification.

The aim of this study is to face this gap by creating and testing an ERC-1155-based POAP attendance badge system. The following research questions are guiding the study:

- **RQ1:** Can an ERC-1155-based smart contract reliably support POAP event creation, single badge issuance, batch badge issuance, and attendee badge verification?
- **RQ2:** Can role-based access control prevent unauthorized users from creating POAP events, issuing attendance badges, and managing issuer permissions?
- **RQ3:** What are the functional testing, gas usage, and code coverage results of the implemented POAP smart contract when evaluated using Hardhat?

To address these questions the system was implemented and tested with automated smart contract tests, gas usage analysis, code coverage analysis and by deployment on the Sepolia testnet. This paper continues with the following organization: The Literature Review section elaborates on the existing research regarding POAP, NFT event management, NFT education, and blockchain credential verification. The Methodology segment describes the system architecture, smart contract design, development tools and deployment. The Results and Analysis section displays testing results, gas consumption and code coverage. The Discussion section explains the results and their limitations and the Conclusion summarizes the contribution and any further work.

LITERATURE REVIEW

Dreyfus [1] showed how to implement POAPs in a classroom setting. The study explained that POAPs are non-fungible tokens associated with an event and can be used as proof of participation, incentives, or rewards. This paper is directly relevant as the present project also leverages on POAP badges as the digital proof of event attendance. In the context of event management, Calderone [2] discussed the possibilities of NFTs in sports ticketing and fan engagement. POAPs, ERC-1155, NFT tickets, layer-two solutions, and decentralized identity were covered in the paper. This aligns with the notion that NFTs have the potential to enhance event attendance data and offer other benefits post-event.

Sriram et al. [3] introduced the blockchain-based system for event management with the Sui blockchain. They are exploring ticket fraud, data security, access control, cost of a transaction, and NFT-based event ticketing. While their implementation was based on Sui, not Ethereum, it still had significance in the context of using blockchain for secure event management and access control. Wu and Liu [4] have done a literature review on the use of NFTs in education. They found that applications for NFTs encompass micro-certificates, transcripts, learning experiences, registrations, and rewards for learning. This helps to ensure the educational value of NFT-based attendance badges, as POAP badges can signify student engagement, learning journey, and event accomplishment.

Azkaaraihlahardi and Jaya [5] conducted a research on NFTs and ERC-1155 standard for digital certificate verification. They are valuable as they explain ERC-1155, which is the protocol to create multiple certificates and which offers batch minting. It's a good fit for the technical choice of multiple kinds of event badges in the current project and is in line with what the Type 2 group would like to see in the new badge design. Khati et al. [6] suggested a student certificate sharing system using Blockchain and NFTs. They employed NFTs as certificate ownership, IPFS metadata storage, MetaMask authentication, hashing and digital signatures. This is important since it demonstrates the potential of blockchain to give ownership, provenance and verification for digital credentials.

Mujezinović et al. [7] presented the blockchain based platform for tracking students' achievements with NFT tokens. They built a prototype system to issue NFT badges for coursework, extracurricular activities and skills from universities. Helps to validate participation and achievement in the academic setting by using digital badges. Ge et al. [8] proposed a decentralized certificate verification system based on blockchain and NFTs. Their system was based on Polygon, IPFS, smart contracts, and performance testing to ensure the authenticity of credentials. The study is significant as it demonstrates the potential of blockchain verification to reduce the need for centralized entities and enhance trust in credentials.

The findings from the analysed studies suggest that blockchain and NFTs have huge possibilities in the fields of attendance, education records, digital credentials and achievement tracking in events. While several existing pieces focus on student records or academic certificates, or ticketing on NFTs, the focus of many is on a lightweight POAP attendance badge system. Some other systems also emphasize ERC-721, which is not as effective in managing multiple badge types as ERC-1155. To address this, this project proposes a system that

leverages the ERC-1155 protocol and incorporates a POAP system, together with a role management system for the POAP issuer, batch issuance, and smart contract-based attendance verification.

METHODOLOGY

This paper explains the method used to develop a POAP attendance verification system. The system uses Ethereum smart contracts to issue digital attendance badges to participants. The smart contract was developed using Solidity and the ERC-1155 token standard. Each event is represented by a unique token ID, and attendees can receive badges as proof that they joined the event. The frontend also supports QR-based check in, where attendees can scan a generated event link to claim their badge. The system was tested using Hardhat and deployed to the Sepolia testnet. The testing focused on checking whether event creation, badge issuance, batch issuance, role management, QR-based badge claiming, and badge verification worked correctly.

System Design Overview

The proposed system is a system for verifying attendance based on a blockchain. The smart contract is the primary backend interface for creating event badges, issuance of POAP badges, permission of issuers and verifying the ownership of badges. Users can use the frontend interface to interact with the smart contract deployed, as well as connect their MetaMask wallet. There are three actors in the system: the contract owner, the authorized issuer, and attendee. The contract owner is responsible for the management of the overall system, for defining types of event badges, for updating the status of the events and for granting or denying permission to the issuer. The authorized issuer can issue attendance badges upon the owner's approval. The organizer can also generate an event check-in QR code, which allows attendees to open a claim page and request the badge using their wallet address. The POAP badge will be rewarded to the attendee in their wallet as a proof of attendance.

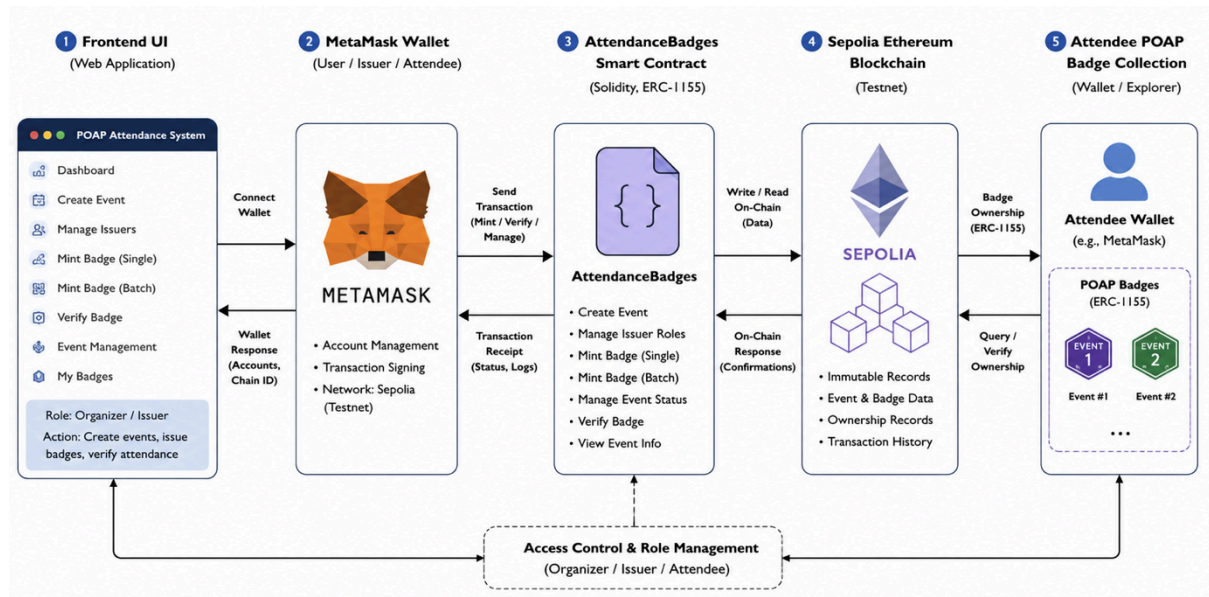


FIGURE 1. System architecture of the proposed POAP attendance verification system.

The workflow starts with the creation of a new POAP event badge type by the owner. The badge type stores a metadata URI, maximum supply, current supply, and active status. Once the event badge has been created, the badge can be issued directly by the owner or an authorized issuer. The batch issuance function allows multiple attendees to receive the same badge in a single transaction. In addition, the frontend provides a QR-based check-in link that attendees can scan to claim the event badge. The claim request is validated before the badge is issued on-chain. After issuance, the attendee can verify badge ownership through the smart contract read functions.

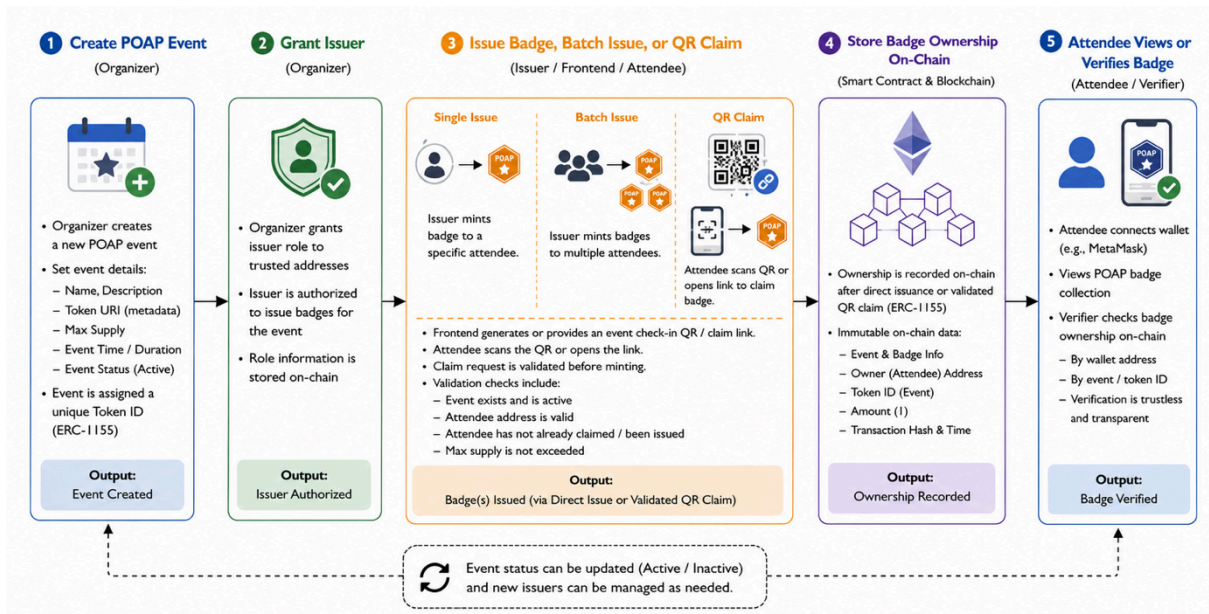


FIGURE 2. Workflow of POAP badge creation, issuance, and verification.

Smart Contract Development

The smart contract is built using Solidity and follows the ERC-1155 multi-token standard. ERC-1155 was selected because it allows multiple event badge types to be managed using different token IDs within a single smart contract. This is suitable for an attendance verification system because different events can be represented without deploying a separate contract for each event. The contract supports event creation, badge issuance, batch issuance, issuer role management, event status management, badge ownership verification, and holder badge queries.

TABLE 1. Summary of main smart contract functions

Function	Purpose	Access Control	Main Validation
createPOAPEvent()	Creates a new attendance badge type for an event.	Owner	Valid metadata URI, max supply, and event time window.
issuePOAP()	Issues one badge to a single attendee.	Owner / Authorized Issuer	Event exists, event is active, valid attendee address, no duplicate badge, and supply limit not exceeded.
batchIssuePOAPs()	Issues badges to multiple attendees in one transaction.	Owner / Authorized Issuer	Applies issuance validation to multiple attendee addresses.
grantIssuer() / revokeIssuer()	Adds or removes trusted badge issuers.	Owner	Valid issuer address and issuer role status.
setPOAPEventActive()	Activates or deactivates an event badge.	Owner	Event token ID must exist.
hasPOAP()	Checks whether an attendee owns a badge for an event.	Public	Valid attendee address and token ID.
getPOAPEvent()	Retrieves event badge information.	Public	Event token ID must exist.

Several validation controls are implemented in the contract to prevent invalid badge issuance. The contract rejects empty metadata URIs, zero-address inputs, invalid event time windows, unauthorized callers, missing token IDs, duplicate badge issuance, inactive event issuance, and issuance beyond the maximum supply. These checks ensure that attendance badges are only issued under valid conditions and that badge ownership can be verified reliably.

Technologies Used

The project was implemented using Solidity, Hardhat, TypeScript, Mocha, Chai, MetaMask, Sepolia testnet, and a simple Next.js/React frontend. Solidity was used to develop the smart contract. Compilation, testing, deployment, gas analysis and code coverage were done using hardhat. The automated tests were in Mocha and Chai. MetaMask wallet was employed in connecting and signing transactions. The public testnet, Sepolia, is chosen to be the location for deploying Ethereum.

Implementation Details and Deployment

Each POAP event is created as a unique ERC-1155 token ID. The smart contract stores event-related information such as the token URI, maximum supply, current supply, event state, and event validity period. When a badge is issued, the contract checks several conditions before minting. These include verifying that the caller has the required issuer authorization, the event exists, the event is active, the attendee address is valid, the attendee has not already received the badge, and the maximum supply has not been reached.

The batch issuance function enhances the usability for more events. The organiser can give the same badge to several of the participants in one transaction, rather than giving out one badge each time. That minimizes the number of repetitive manual transactions and enhances the user experience to the event organizers.

The frontend also has a QR code based check-in system. For each event, the organizer can display a check-in QR code that contains a claim link for the event. System verifies if the event ID, wallet address, QR check-in code is valid, if the event is active, and if the attendee has already redeemed the check-in badge. If the validation passes, the badge is issued to the attendee through the smart contract and the transaction can be viewed on Sepolia.

The AttendanceBadges smart contract was deployed to the Ethereum Sepolia testnet using Hardhat. As part of the deployment process, the contract address, deployment transaction hash, ABI file, and deployment metadata were generated. The deployment was completed on the Sepolia network with chain ID 11155111. The deployed contract address is 0xb4342aB9c0035FED4a3b25e9455e2106C4D97E24, and the deployment transaction hash is 0xf4e215f3c87a65d4dfb2c6a7810a1a6bb8dbe2702d4c2fb7baf46c5efc059b1b.

RESULTS AND ANALYSIS

Testing Outcomes

The functional testing was performed with Hardhat, Mocha and Chai. The automated test suite included deployment, creation of events, issuing of badges, batch issuance, role management, access control, validation checks and read functions. 15 automated test groups passed with success.

TABLE 2. Functional testing summary

Test Category	Test Focus	Result
Deployment	Owner, issuer role, ERC-1155 interface, constructor validation	Passed
Event Creation	Valid event creation, invalid input rejection, owner-only restriction	Passed
Badge Issuance	Single issuance, authorized issuer issuance, duplicate rejection	Passed
Batch Issuance	Multiple attendees, empty array rejection, zero address rejection	Passed
Admin Roles	Grant issuer, revoke issuer, invalid role input rejection	Passed
Read Validation	Missing token ID and zero address validation	Passed
Overall	15 automated test groups	Passed

The test results demonstrate that the contract is able to process successful and unsuccessful operations. Badges can be created and issued by valid owners and issuers, and invalid inputs and users are not accepted. The contract also ensures that there will be no duplicates, no inactive events, and no overflow of maximum supply.

Performance Metrics

The transaction cost of important smart contract operations was measured through gas analysis. A total of 2,563,405 gas was used for contract deployment. The createPOAPEvent function had an average gas cost of 170,000, and the issuePOAP function had an average gas cost of 179,584. It took 381,015 gas to execute the batchIssuePOAPs function.

TABLE 3. Gas usage summary

Function	Average Gas	Calls
Deployment	2,563,405	14
createPOAPEvent	170,000	10
issuePOAP	179,584	4
batchIssuePOAPs	381,015	1
grantIssuer	48,691	2
revokeIssuer	26,651	1
setPOAPEventActive	27,803	2

Additionally, code coverage analysis was carried out. The contracts/AttendanceBadges.sol file was 100.00% covered at the line level and 100.00% covered at the statement level. This means that the primary contract logic, validation rules and paths to revert were tested during automated testing.

Key Findings

The results presented above demonstrate that the proposed POAP system is able to help with verification of event attendance using a blockchain. All automated tests were passed and the core smart contract functions acted as expected, indicating that the system as a whole passed. Access control by role did not allow badges to be issued incorrectly. Maximum supply check to prevent exceeding event's supply limit. The duplicate issuance check ensured that the same attendee could not get more than one POAP. Although batch issuance consumed more gas than single issuance, it is more practical for events with many attendees because multiple badges can be issued in one transaction. This decreases duplicate manual transactions and enhances the user-friendliness of the event organizers.

DISCUSSION

The results indicate that the proposed solution is able to solve the problem of weak attendance verification in centralized system. A central admin may change the traditional attendance records, while the proposed system records the ownership of the badge on-chain. This is more transparent since ownership of the badge can be checked by reading the blockchain records.

This project uses ERC-1155 as an appropriate use of smart contracts to handle multiple event badge types. This approach is more convenient than deploying a separate contract for each event. The batch issuance function will also enhance system usability as it allows for the realistic case of issuing many badges at the end of an event.

The security design also assists in impacting the project objective. Owner-only functions do not allow for the creation of event badges or changes to the status of an event by ordinary users. Issuer role management enables trusted users to issue badges without providing complete control over the badge as an owner. Validation checks block creation of badges with zero address, duplicate badges, badges without a token, issuing an inactive event, and exceeding the maximum supply.

At the same time, there are some restrictions. Firstly, Sepolia is a testnet, and deploying on the Ethereum mainnet will require actual fees and might necessitate further gas optimization. Second, blockchain-based badges establish ownership of a wallet address, but don't automatically validate the real-world identity of the attendee. Thirdly, the front end user interface may be enhanced by adding a more comprehensive event dashboard, user-friendly error messages, and a more detailed user interaction. Fourth, Badge images and event details can be stored on IPFS, enhancing metadata storage.

The system shows potential for practical use. It may be used for university events, workshops, seminars, conferences, training programs and professional seminars. Participants can carry an attendance certificate with them and the organizers will decrease the need for manual attendance sheets. This enhances the level of trust from organizers, participants, and third party verifiers.

CONCLUSION

This project proposed a blockchain-based POAP attendance badge system for event attendance verification. The system was implemented using Solidity, ERC-1155, Hardhat, MetaMask, and the Sepolia testnet. It allows event organizers to create event badge types, manage authorized issuers, issue badges to individual attendees, issue badges in batches, and verify whether an attendee owns a valid badge. The evaluation showed that all 15 automated test groups passed successfully. Gas usage analysis was conducted to measure the transaction cost of the main smart contract functions, while code coverage analysis showed 100% line and statement coverage for the smart contract. These results indicate that the system works correctly as a prototype for blockchain-based attendance verification.

However, the system still has several limitations. It was deployed on the Sepolia testnet rather than the Ethereum mainnet, and badge ownership only proves ownership of a wallet address, not the real-world identity of the attendee. Future improvements may include stronger identity verification, IPFS-based metadata storage, gas optimization, and a more complete frontend dashboard for event organizers and attendees.

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